

Sirbu Mihai-Alexandru, Sacasanu Victor-Andrei, Enache David-Alexandru //// 1231EC-IoT

FlowState

A real-time, shared environment for collaborative productivity and deep focus.

The Vision

Modern studying is often isolated and distracting. FlowState bridges the gap between solo productivity and social connection.

Goal: Create a digital "third space" that encourages accountability through visual presence.



CREATIVE ROOTS

Inspiration: Virtual Cottage

Study

60 Minutes to go



The Technical Pillars



Real Auth

Firebase-backed security
for data persistence
across any device.



State Logic

BLoC pattern for
predictable, complex
timer transitions.



Real-time

WebSockets on Render for
sub-second
synchronization.

Real Authentication

FIREBASE AUTH

Secure & Persistent

Moving beyond anonymous sessions to allow users to access their focus history and profile from any hardware.

- Cross-device data synchronization.
- Unique User Identifiers (UID).
- Email/Password credential management.

Firestore Implementation

- Firebase Authentication
- Cloud Firestore (We used it to create a specific "vault" for each user based on their UID)
- Real-Time Data Streams (Reactive UI)

 (default)	 users  	 6Nsmp2T0f1gPGdyqdAeTmCRFRX72
+ Start collection	+ Add document	+ Start collection
users >	6Nsmp2T0f1gPGdyqdAeTmCRFRX72 >	+ Add field
		displayName: "user1" totalStudySeconds: 180

 Search by email address, phone number, or user UID

Add user  

Identifier	Providers	Created ↓	Signed In	User UID
mihai.s@gmail.com		May 12, 2026	May 12, 2026	p9MqGKORRPW56dqEln4Rxe...
victor.s@yahoo.com		May 12, 2026	May 12, 2026	jbXjnER9fRMF9OjiOzi9fW01v...
hamham@meow.com		May 12, 2026	May 12, 2026	4iKjQhoKE9PMM0hWF19k0w...
howhow@meow.com		May 12, 2026	May 12, 2026	BYE62pLLGvcqQSI0XKzBfVU...
sirbu.m.alex@gmail.com		May 11, 2026	May 12, 2026	6Nsmp2T0f1gPGdyqdAeTmC...

State Management



The Timer Engine

Using the Business Logic Component (BLoC) pattern to decouple UI from the Pomodoro logic.

Ensures the countdown remains accurate even during network jitter or background transitions.



←

FlowState — Room #67664

DEBUG

Session Tasks

Add a task...

+

☐ math exam

☐ study ww2

FOCUSING

09:47

Total Progress: 00:13 / 10:00

⏸

Who's Here

● user1 (You) 🏰

Studying

● Enache David

Studying

The Server

Hosted on **Render**, our Node.js WebSocket server acts as the source of truth,
synchronizing study phases across all connected participants instantly.

room states

- waiting
- active
- ended

phase states

- study
- break
- null

native websocket states

- broadcast_to_room
- broadcast_room_send_to_user

user states

- is_ready
- personal_break_started_at
- personal_break_paused_for_scheduled_
break

The User Journey

Authentication	The Lobby	Syncing	Focus
Secure login/sign-up via Firebase.	Enter the lobby to Host or Join.	Gets in-sync with the other users	Deep work in the study room.

Future Roadmap



Leaderboards

Global and Friend-specific rankings based on total hours of deep work logged. Gamifying productivity.



Social System

Persistent Friend Lists to quickly see who is online and instantly invite them to private study rooms.



Lo-Fi Radio

Built-in curated study beats and support for custom, user-inputted audio queues to maintain the vibe.

FLOWSTATE TEAM

That's it

<https://github.com/better-call-sirbu/FlowState>